Counseling Module

Lecture 5: Core AI

Prof. Dr. Michael Granitzer

October 15, 2025

AI-BSC-Coun-CoreAI

- •Studying at the University of Passau –
- •B.Sc. Artificial Intelligence

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- •Study Programme from the AI perspective
- •How do modules apply to modern AI Systems
- •Core AI Modules in detail
- •Elective Modules in (more) detail
- •Questions & Answers

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Today’s Frontier AI Method - Transformers

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Transformer Architecture – Attention + Deep Learning

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Theoretical Analysis of Transformers / AI Models

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- •How to implement such a Transfomer Model?
- •Can we keep it modular and componets re-usable?
- •What data and programme structure gives us best efficiency?
- •Deploying and operating such a model?
- •How to engineer the data?
- •How to scale such systems

Realising Transformer/AI Models

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Evaluating & Applying Transformer/AI Models

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- •How to evaluate AI Systems?
- •How to adapt it to different modalities?
- •How to build Agent systems on top of AI models?
- •Impact of AI systems and how to regulate them?
- •Wider application of AI Systems
- •…

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Machine Learning

Topics

- •History of AI, basic concepts & sub-areas
- •Basics of machine learning (supervised and unsupervised learning, features)
- •Association analysis
- •Cluster analysis
- •Classification algorithms
- •Regression methods
- •Evaluation methods for learnt models

Prof. Herbold, AI Engineering

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Probabilistic Machine Learning

Topics

- •Fundamentals of Statistical Learning and Probabilistic Machine Learning, Learning as Inference
- •Data Representation / Feature Engineering
- •Models: Naive Bayes, Probabilistic Graphical Models, Bayesian Belief Networks, Bayesian Linear Models, Mixture models
- •Model Evaluation, Model Selection, and Hyperparameter Tuning

Prof. Granitzer, Data Science

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Deep Learning

Topics

- •Basics of Representation Learning
- •Perceptron LearningFeedforward Neural Networks
- •Gradient Descent and Backpropagation
- •Regularization in Deep Learning
- •Advanced Neural Network Architectures such as Convolutional and Recurrent Neural Networks
- •State-of-the-art developments in Deep Learning

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Multiagent Systems

Topics

- •Distributed Optimization, Nash-Equilibrium, Congestion Games, Pricing,
- •Resource Allocation, Auctions, Mechanism Design, Learning in Games

You are able to model multiagent distributed systems with strategically interacting agents and algorithmically solve such systems by means of computing equilibrium solutions.

You learn the theory of mechanism design for distributed resource allocation problems

Prof. Harks, Mathematical Optimisation

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AI Project, AI Seminar & Bachelor Thesis

AI-Project

Solve a (more) complex AI problems / topics as team

Responsible: Granitzer, Harks, Herbold, Lemmerich

AI-Seminar

Individual work on deepening knowledge in a specific topic, write it up, present it and discuss with colleagues

Responsible: every professor

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Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

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Elective Modules

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You choose through

- •16 ECTS from the chosen compulsory elective group
- •45 ECTS from three elective groups
-   - –Each elective group must comprise at least 10 ECTS;
- –At least one elective group must come from Computer Science or Mathematics.
- –German as a Foreign Language with at least 20 ECTS

(recommended, if you want to establish yourself in Germany)

- •Details in the Module Catalog:

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Compulsory Electivse – Advaced AI Applied

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Compulsory Electives – Advanced AI Theory

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Elective Modules – Mathematics

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- •https://www.menti.com/al1m3pzeozro

Questions and Answers

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Any Questions?

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